

Natural flavonoid morin showed anti-bacterial activity against *Vibrio cholera* after binding with cell division protein FtsA near ATP binding site

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ABSTRACT

Background: Increasing antibiotic-resistance in bacterial strains has boosted the need to find new targets for drug delivery. FtsA, a major bacterial divisome protein can be a potent novel drug-target.

Methods and results: This study finds, morin (3,5,7,2',4'-pentahydroxyflavone), a bio-available flavonoid, had anti-bacterial activities against *Vibrio cholerae*, IC₅₀ (50 µM) and MIC (150 µM). Morin (2 mM) kills ~20% of human lung fibroblast (WI38) and human intestinal epithelial (HIEC-6) cells in 24 h *in-vitro*. Fluorescence studies showed morin binds to VcFtsA (FtsA of *V. cholerae*) with a K_d of 4.68 ± 0.4 µM, inhibiting the protein's polymerization by 72 ± 7% at 25 µM concentration. Morin also affected VcFtsA's ATPase activity, recording ~80% reduction at 20 µM concentration. The *in-silico* binding study indicated binding sites of morin and ATP on VcFtsA had overlapping amino acids. Mant-ATP, a fluorescent ATP-derivative, showed increased fluorescence on binding to VcFtsA in absence of morin, but in its presence, Mant-ATP fluorescence decreased. VcFtsA-S40A mutant protein did not bind to morin.

Conclusions: VcFtsA–morin interaction inhibits the polymerization of the protein by affecting its ATPase activity. The destabilization of VcFtsA assembly in-turn affected the cell division in *V. cholerae*, yielding an elongated morphology.

General significance: Collectively, these findings explore the anti-bacterial effect of morin on *V. cholerae* cells targeting VcFtsA, encouraging it to become a potent anti-bacterial agent. Low cytotoxicity of morin against human cells (host) is therapeutically advantageous. This study will also help in synthesizing novel derivatives that can target VcFtsA more efficiently.

1. Introduction

Cholera, caused by a gram-negative bacterium, *Vibrio cholerae*, is an acute infectious disease, characterized by profuse watery diarrhea. According to WHO reports (WHO media center report on Cholera), there are many serogroups of *V. cholerae*, but of them only two, O1 and O139, cause outbreaks across the globe. Previously, only *V. cholerae* serogroup O1 was considered to be the cause of epidemic outbreaks. However, in 1993, a new serogroup designated as *V. cholerae* serogroup O139 was first isolated from the Indian subcontinent region [1]. By routine treatment of a community with broad-spectrum antibiotics, or mass chemoprophylaxis, little effect was achieved on the control of the spread

of cholera. This was due to the rapid surfacing of resistant strains [2]. The emergence of resistance against various antibiotics in *V. cholerae* is the major problem faced nowadays. It is thus necessary to find some potential drugs that can bind novel targets thereby blocking the spread of cholera with prolonging sensitivity.

FtsA is one of the major proteins present in the cell divisome complex of bacteria. It functions by anchoring the Z-ring (formed by FtsZ) to the cell membrane at mid-cell and facilitates the Z-ring constriction leading to effective septum formation during cytokinesis phase of cell division in bacteria [3]. The proteins of the divisome complex are potent antimicrobial targets because the hampering of cell division stalls the spread of virulent bacterial strains. In the past, few studies have aimed at

Abbreviations: VcFtsA, FtsA of *Vibrio cholerae*; VcFtsZ, FtsZ of *Vibrio cholerae*; ATP, Adenosine Triphosphate; PIPES, piperazine-N,N'-bis(2-ethanesulfonic acid); IPTG, isopropyl-β-D-thiogalactopyranoside; ATP, adenosine 5'-triphosphate; Ni²⁺-NTA, Ni²⁺-nitrilotriacetic acid; BSA, bovine serum albumin.

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targeting FtsZ (another major divisome protein) by screening potential small molecules [4–6]. This lacks in the case of FtsA. The in-detail study of FtsA from *Vibrio cholerae* (VcFtsA) [7] from our group has led us to think that this protein can be a novel target for new-generation drugs to control the spread of *V. cholerae*.

Morin (3,5,7,2',4'-pentahydroxyflavone), a yellow natural bioflavonoid, is a major component of traditional medicinal plants like almond (*Prunus dulcis*), fig (*Chlorophora tinctoria*) and also other members of Moraceae family fruits like Indian guava (*Psidium guajava*). Morin and its derivatives are known as anti-microbial agents [8–10]. On the other hand, morin has low cytotoxicity [11,12]. Reports have also mentioned the protective effects of the flavonoid against chemotoxicity [11]. All these properties made morin a potent candidate, which can be tested for its anti-microbial effect targeting the divisome complex proteins. Taking advantage of the fact that there are no previous studies of morin and its effect on cell division proteins, we got interested in elucidating the mechanism of action of morin on *V. cholerae* N16961 cells. A comprehensive study on the effect of morin on VcFtsA has been made by us here. The binding parameters of the interaction and studying the nature of the binding site will help us in the future to synthesize new derivatives of morin that will have higher potency in targeting VcFtsA hence showing better efficacy in controlling the spread of *Vibrio cholerae*.

2. Materials and methods

2.1. Bacterial strains

E. coli host strain BL21(DE3)/pLysS (Novagen, USA) was used for the regulated expression of *V. cholerae* FtsA and *V. cholerae* FtsZ. *Vibrio cholerae* O1 biovar El Tor str. N16961 (Non-virulent serotype) was used for demonstration of the effect of morin on *V. cholerae*.

2.2. Animal cell culture

Normal human lung fibroblast (WI38) and cells were cultured in nutrient mixture DMEM supplemented with 10% FBS, 3.7 g/l sodium bicarbonate, and a 1% antibacterial and antimycotic solution containing penicillin, streptomycin, and amphotericin B. On the other hand, normal human intestinal epithelial (HIEC-6) cells were cultured in Opti-MEM™ I Reduced Serum Medium (Thermo, USA). Cells were grown in a humidified atmosphere of 5% carbon dioxide and 95% air in an incubator at 37 °C.

2.3. Preparation of morin stock

Morin hydrate (morin) purchased in powder form from Sigma, USA, was dissolved in 100% DMSO and stored at –20 °C. The final concentrations of DMSO in all the experiments were kept at 0.05%.

2.4. Growth curve

To determine the growth curve of *V. cholerae* N16961 in the presence of morin, the cells were cultured in LB media overnight at 37 °C. Growth kinetics was performed in 96-well microtiter plates by adding 2% inoculum (v/v) from overnight cultures in 100 µl LB media, either in the absence or in the presence of different concentrations of morin at 37 °C. The optical density (OD) of the culture at 590 nm was monitored at regular time intervals for 16 h with a Micro-plate reader (Multiskan EX, Labsystems). Each set of experiments was repeated three times.

2.5. Cytotoxicity assay

The growth inhibitory effects of morin on normal human cell lines were assessed by the MTT assay [13,14]. In brief, the cultured WI38 and HIEC-6 cells in their logarithmic phase were seeded into a 96-well microtiter plate (1×10^4 cells/ml) and treated with morin in varying

concentrations for 24 h. After treatment, 50 µl of the MTT solution (3 mg/ml) was added to each well and incubated for 1 h at 37 °C followed by the dissolution of the formazan crystals formed in 50 µl of DMSO. Absorbance was measured at 570 nm with an enzyme-linked immunosorbent assay reader (Multiskan EX, Labsystems). Data obtained were represented as the percentage of cell death by the following formula;

$$\% \text{Cell Death} = \left[\left[1 - \left(\frac{A_T}{A_C} \right) \right] \times 100 \right] \%$$

where, A_T and A_C are the absorbances obtained from the test samples and the solvent control respectively [15–17].

2.6. Visualization of cell morphology

V. cholerae N16961 cells were treated with doses of morin (30 µM and 60 µM) in LB medium and incubated at 37 °C for 4 h. Cells were centrifuged and resuspended in PBS (pH 7.4). Next, the cells were fixed by a mixture of 2.8% formaldehyde and 4% glutaraldehyde solution in PBS (pH 7.4). Cells were stained with DAPI (nuclear DNA staining dye) and examined using Olympus LSM IX81 confocal microscope at 100× under the differential interference contrast mode. Images obtained were analyzed with Olympus Fluoview. Morphology of WI38 and HIEC-6 cells were examined with a Bright field microscope under 20× objective.

2.7. Purification of VcFtsA and VcFtsZ

His-tagged VcFtsA [7] and VcFtsZ [18] were purified according to the previously reported work done in our laboratory by our group [7,18]. Affinity column chromatography using Ni^{2+} -NTA agarose column was used to purify over-expressed His-tagged VcFtsA [7] and VcFtsZ [18] separately using imidazole gradient. His-tags of both the proteins were thereafter removed by digestion with 2 units/ml thrombin (GE Healthcare, USA) on ice. Thrombin was removed post-digestion in a pre-equilibrated Benzamidine-Sepharose column (GE Healthcare, USA). Both the purified proteins, VcFtsA (thrombin digested) and VcFtsZ (thrombin digested), were dialyzed separately against the same storage buffer (25 mM PIPES (pH 7.0), 1 mM DTT and 10% glycerol) and stored at –80 °C. Before the respective experiments, VcFtsA or VcFtsZ was dialyzed in the suitable buffers at 4 °C. Protein concentrations of separately purified VcFtsZ and VcFtsA were measured by Bradford method [19], taking BSA as a standard using a spectrophotometer.

2.8. Light scattering assay

Polymerization of VcFtsA (6 µM) or VcFtsZ (6 µM) was monitored by measuring 90° light scattering in a fluorimeter (Photon Technology International, USA, Model QuantaMaster™) equipped with a temperature-controlled cuvette holder assembly. VcFtsA was incubated with a polymerization buffer (25 mM PIPES, 10 mM MgCl_2 , 5 mM CaCl_2 , 1 mM ATP, pH 7.0) at 37 °C, whereas VcFtsZ was separately incubated with another polymerization buffer (25 mM PIPES, 10 mM MgCl_2 , 10 mM CaCl_2 , 1 mM GTP, pH 7.0) at 37 °C. Assembly kinetics of both VcFtsA and VcFtsZ were monitored separately by a fluorimeter keeping excitation and emission wavelengths at 550 nm.

2.9. Measurement of ATPase activity of VcFtsA

The standard malachite green-ammonium molybdate assay having some modifications was used for this purpose [20,21]. VcFtsA (6 µM) was polymerized in polymerization buffer (25 mM PIPES, 10 mM MgCl_2 , 5 mM CaCl_2 , 1 mM ATP, pH 7.0) at 37 °C in the presence of different concentrations of morin (0–20 µM) for various time intervals (0–105 min). Reactions were quenched with 10% (v/v) perchloric acid at desired time intervals. The ATPase activity of VcFtsA was monitored by measuring the amount of inorganic phosphate (Pi) released at different

time-points as described [7]. Moles of Pi released per Mole of VcFtsA were plotted against time and the rate was calculated from the slope of the linear fit using GraphPad Prism software.

2.10. Pectral measurements

All fluorescence measurements were performed using Photon Technology International, USA, Model QM-4CW, which is equipped with a Peltier system for temperature control and a quartz cuvette of 1 cm path length was used. Inner filter effect correction was done by the formula;

$$F = [F_{obs}] \times \left[\text{antilog} \left[\frac{(A_{ex} + A_{em})}{2} \right] \right]$$

where, F_{obs} is the observed fluorescence, A_{ex} is the absorbance at the excitation wavelength and A_{em} is the absorbance at the emission wavelength [22]. All absorbance measurements were performed using a quartz cuvette of 1 cm path length in a JASCO V-630 UV-visible spectrophotometer. The far-UV CD spectra were recorded using a 0.1 cm path length cuvette in a JASCO J600 spectropolarimeter at 30 °C over the range of 260–200 nm wavelength.

2.11. Job plot for VcFtsA-Morin interaction

Using the continuous variation method, VcFtsA – morin binding stoichiometry was determined [23]. VcFtsA and morin were mixed in varying concentrations in several mixtures keeping the total concentration of VcFtsA and morin constant at 15 μ M. The samples were excited at 295 nm wavelength and the change of tryptophan fluorescence was then recorded at 330 nm wavelength at 30 °C.

2.12. Binding parameters of VcFtsA – Morin interaction

VcFtsA – morin binding parameters were obtained from both the double-reciprocal plot [24] and Scatchard plot analysis [25–27]. The intrinsic tryptophan fluorescence of 1 μ M VcFtsA was recorded in the presence of an increasing concentration of the morin ligand (0–30 μ M). Change in tryptophan fluorescence was recorded at 330 nm wavelength at 30 °C by exciting the samples at 295 nm. The change in tryptophan fluorescence was plotted against the respective concentrations of morin and later fitted to a single binding site quadratic equation from which the dissociation constant (K_d) was obtained in GraphPad Prism software;

$$AB = \frac{(A_T + B_T + K_d) - \sqrt{(A_T + B_T + K_d)^2 - 4(A_T B_T)}}{2}$$

where, AB denotes the change in fluorescence signal, A_T is the concentration of VcFtsA in the reaction mixture, B_T is the concentration of morin titrated into VcFtsA and K_d is the dissociation constant of the interaction.

To determine the dissociation constant (K_d) from the Double-Reciprocal plot, the following relationship was used;

$$\frac{1}{X} = 1 + \frac{K_d}{[L_f]}$$

where, $[L_f]$ represents free morin concentration, X is the fraction of binding sites occupied by morin and K_d is the dissociation constant. X was determined from the relation $X = (F - F_0) / F_{max}$, where F_0 is the fluorescence intensity of VcFtsA in absence of morin, F is the corrected fluorescence intensity of VcFtsA in the presence of different concentrations of morin, and F_{max} is calculated from the plot $1 / (F - F_0)$ versus $1 / [\text{Morin}]$ and extrapolating $1 / [\text{Morin}]$ to zero. $[L_f]$ was determined from the equation, $[L_f] = C - X[Y]$, where C is the total concentration of morin and [Y] is the molar concentrations of ligand binding sites using a stoichiometry of 1:1 as determined from the Job Plot.

For Scatchard plot, the following formula was used;

$$\frac{r}{[L_f]} = nK_a - rK_a$$

where, K_a represents the association constant, the ratio of the concentration of bound ligand to the total protein is denoted by r, and n is the maximum number of ligand (Morin) binding sites per VcFtsA molecule. The K_a is determined from the slope of the plot of $r/[L_f]$ against r, and the 'n' is obtained from the extrapolated value of r at $r/[L_f]$ equals to zero [27].

The enthalpy (ΔH) of VcFtsA-Morin interaction was calculated from the Van't Hoff equation by plotting $\ln K_a$ at different temperatures against $(1/T)$ assuming that the enthalpy change is constant throughout temperature changes [28]. The following equation was used;

$$\ln K_a = -\frac{\Delta H}{RT} + \frac{\Delta S}{R}$$

where, R is the gas constant and ΔS is the entropy of the reaction. The ΔG (change in Gibb's free energy) was calculated at 30 °C from the standard equation;

$$\Delta G = -RT \ln K_a$$

ΔS at 30 °C was calculated from the following equation;

$$\Delta G = \Delta H - T\Delta S$$

All curve fits and calculations were done in GraphPad Prism software.

2.13. Docking studies

2.13.1. Homology modeling of VcFtsA

At first, the 3D structure of VcFtsA having the following primary amino acid sequence was modelled using the SWISS-MODEL workspace.

MTKTVDDNIHVLGIDIGTATVSALVGEILPDGQINII-
GAGSSPSRGMDKGGVNDLESVVKSVQRAIDQAEMMAECK-
ISHVFLSISGRHIASRIKMGITISDEEVSQEDM-
DRAIHTAKSIKIGEEQRILHVIPQEFTIDYQE-
GIKNPLGLSGVRMEVSVHLITCHNDMARNIIKAVERCGLKVEQLVYS-
GLAASNAVITEDER-
ELGVCVVDIGAGTMDIAIWTGGALRHTEVFSYAGNAVTSIDIA-
FAFGTPVSDAEIIVKYGICALSELVSKDDTVNVPSVGGPRSRSLRQRTLSE-
VIEPRYTELMGLVNQTVDSVQESLRKEGIKHHLAAGVVLTTGGAA-
QIEGLVACAERVFRNQVRVGKPLEVSGSLTDYVKEPYH-
STAVGLLHYARDSQISDDTEYQEPKRPSTGTWIGRLRNWIQKEF.

Template search was performed with BLAST and HHblits against the SWISS-MODEL Template Library (SMTL, last update: 2018-05-23, last included PDB release: 2018-05-18). The target sequence was searched with BLAST against the primary amino acid sequence contained in the SMTL. A total of 9 templates were found. An initial HHblits profile was built using the procedure outlined in [29], followed by 1 iteration of HHblits against the NR20 database. The obtained profile was then searched against all profiles of the SMTL. A total of 813 templates were found. For each identified template, the template's quality was predicted from features of the target-template alignment. The templates with the highest quality were then selected for model building. Models were built based on the target-template alignment using ProMod3. Coordinates that are conserved between the target and the template are copied from the template to the model. Insertions and deletions are remodeled using a fragment library. Side-chains are then rebuilt. Finally, the geometry of the resulting model is regularized by using a force field. In the case of loop modeling with ProMod3 fails, an alternative model is built with PROMOD-II [30]. The global and per-residue model quality has been assessed using the Qualitative Model Energy Analysis (QMEAN) scoring function [31]. For improved performance, weights of the individual QMEAN terms have been trained specifically for SWISS-MODEL. The quaternary structure annotation of the template is used to model the

target sequence in its oligomeric form. The method [32] is based on a supervised machine learning algorithm, Support Vector Machines (SVM), which combines interface conservation, structural clustering, and other template features to provide a Quaternary Structure Quality Estimate (QSQE). The QSQE score is a number between 0 and 1, reflecting the expected accuracy of the inter-chain contacts for a model built based on a given alignment and template. Higher numbers indicate higher reliability. This complements the Global Model Quality Estimation (GMQE) score which estimates the accuracy of the tertiary structure of the resulting model.

2.13.2. In-silico docking of VcFtsA and Morin

In-silico docking studies were performed with AutoDock Vina to find out the putative binding site of morin to the modelled protein [33]. Since the protein was modelled using the template 3wt0.1.A.pdb, we used 3wt0.pdb (crystalline structure of cell division protein ftsA, complexed with ATP and Mg) as the reference molecule. 3wt0.pdb has four identical amino acid chains. The chains B, C, D, and the associated ligands were deleted with Discovery Studio Client (v 19.1.0.18287). The ATP binding site (active site) in chain A was located and the ATP molecule was saved as reference pose. MarvinSketch 16.3.28 was used to draw and generate three dimensional structures of the ligand molecules (morin and ATP). AutoDock Tools 1.5.6 was used to prepare the ligands, modelled protein, and finding out the size of grid. A grid of size 65x65x65 with 1 Å grid spacing was considered around the centre of the protein molecule (−5.717, −1.325, 55.264) to fit the entire protein molecule within the grid. The protein molecule was considered as rigid structure, whereas the ligand had rotatable bonds. The exhaustiveness was set to 8 and the AutoDock Vina was run using AutoDock Tools 1.5.6 interface. The best poses were predicted along with the affinity of binding to specific site of the protein molecule. The Rank 1 pose with highest affinity (kcal/mol) was considered.

The best pose of ATP was bound at a site of modelled protein which is similar to the ATP-binding site of chain A of 3wt0.pdb. The two poses of ATP nearly overlapped with a RMSD value of 8.8 Å. Thus, it was confirmed that this protocol of molecular docking can predict the binding site and pose of small molecule to the modelled protein. Next, morin was docked using the same protocol. Discovery Studio Client (v 19.1.0.18287) was used for molecular graphics and analysis of ligand-protein interactions.

2.13.3. Effect of S40A mutation on morin binding

VcFtsA with S40A mutation was modelled using the same protocol as described above. For docking studies, a grid of size 15x15x15 with 1 Å grid spacing was considered around the 40th amino acid Alanine and docking of Morin was performed at the same pocket it bound to the wild type protein. The exhaustiveness was set to 8 and the AutoDock Vina was run using AutoDock Tools 1.5.6 interface.

2.14. Membrane potential ($\Delta\psi$)

The membrane potential established within *V. cholerae* N16961 cells was determined using the reagents provided in the BacLight™ Bacterial Membrane Potential Kit (Invitrogen, USA) and Fluorescence spectroscopy [34,35]. First, overnight grown *V. cholerae* N16961 cells were inoculated in fresh LB medium and grown for 3 h at 37 °C. Approximately 1×10^5 cells were harvested and resuspended in PBS and further incubated without or with 1 µM of carbonyl cyanide 3-chlorophenylhydrazone (CCCP) or morin (50 µM) at 25 °C for 30 min. 20 µM 3,3'-diethyloxacarbocyanine iodide (DiOC₂(3)) was added to each of the reaction sets after incubation and further incubated for 30 min in dark. Each reaction set was then excited separately at 470 nm and the emission spectra (500–550 nm) were monitored using a fluorescence spectrometer (Photon Technology International, USA, Model QM-4CW) [4]. The corrected spectra were obtained by subtracting the spectra of relative blanks from each data set.

2.15. Site-directed mutagenesis

The standard site-directed mutagenesis technique [36,37] was used for producing the VcFtsA mutant. Primers used are as mentioned in Table 1. Briefly, using PCR method, the plasmid containing the mutant gene was formed and amplified. Subsequent digestion with *DpnI* (New England Biolabs, UK) removed the parent plasmid. Later the mutagenized plasmids were transformed into XL10-Gold *E. coli* cells for ligation.

3. Results

3.1. Effect of morin on the growth of *V. cholerae* N16961 cells

The effect of morin on the growth of *V. cholerae* N16961 cells was described in Fig. 1A. The growth kinetics of *V. cholerae* N16961 cells in the presence of different concentrations of morin (0 µM – 150 µM) clearly indicated that morin could inhibit bacterial growth in a dose-dependent manner (Fig. 1A). The IC₅₀ of morin from the growth curves was observed to be 50 µM. No visible growth was seen at 150 µM concentration of morin. Minimum Inhibitory Concentration (MIC) of an antibacterial agent for a particular bacterium is defined as the lowest concentration, which causes complete inhibition of growth of a bacterial population ($\sim 10^6$ cells per ml) during incubation at 37 °C. From the kill curves (Fig. 1B) it could be confirmed that MIC value for morin was 150 µM. It was also observed that doses higher than MIC could kill *V. cholerae* N16961 cells at rates proportional to the concentrations of morin. 500 µM morin could kill 10^6 cells of *V. cholerae* N16961 in approximately 3 h.

3.2. Effect of morin on mammalian cell lines

The cytotoxic effect of morin on human cells was determined using normal human intestinal epithelial (HIEC-6) cells and normal human lung fibroblast (WI38) by MTT assay. We observed from the MTT assay that morin at 2 mM concentration could only kill about 20% of both WI38 and HIEC-6 cells in 24 h (Fig. 2A). On the other hand, in previous experiment it was observed that 500 µM morin killed 10^6 cells of *V. cholerae* N16961 in approximately 3 h. Therefore, the results indicate that morin was negligibly cytotoxic against normal mammalian cells in comparison to its effect on bacterial cells. The observed morphology of both the HIEC-6 and the WI38 cells did not change on treatment with 1 mM morin for 24 h (Fig. 2B & C). This suggests that morin at this dose did not induce any stress response within the experimental mammalian cells.

3.3. Morin induces elongation of *V. cholerae* N16961 cells

Owing to the fact that morin inhibited the proliferation of *V. cholerae* N16961 cells, we wanted to examine the morphological changes, if any, occurring in the *V. cholerae* N16961 cells when incubated in the presence of morin at 37 °C. For this purpose, the bacterial cells were incubated in the absence or in the presence of 25 µM (half of the IC₅₀) and 50 µM (IC₅₀) concentrations of morin at 37 °C for 4 h. It was observed that morin induced elongation of the *V. cholerae* N16961 cells (Fig. 3). Bacterial cells grown without morin had an average length of 2.26 ± 0.32 µm. At 25 µM morin concentration, the average cell size increased to 4.23 ± 1.21 µm, which is nearly two-fold. The average size of *V. cholerae* N16961 cells increased to about four-fold, 8.86 ± 3.84 µm, in the

Table 1
Forward and Reverse primers used for VcFtsA-S40A mutant protein.

PRIMERS	SEQUENCES
VcFtsA-S40A	
S40A – Forward Primer	5'-GTAACAGTGAGATCGCGTTT-3'
S40A – Reverse Primer	5'-AAACGCGATCTCACTGGTTAC-3'

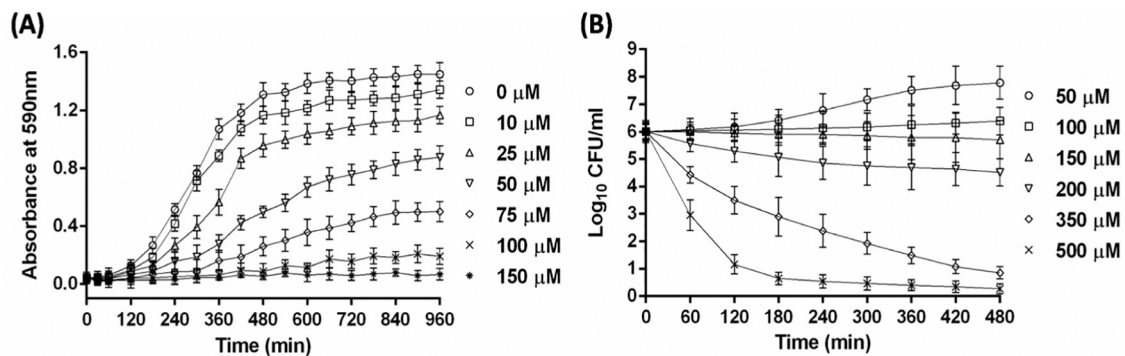


Fig. 1. Growth Curves and Kill Curves. (A) *V. cholerae* N16961 cells grown in the absence or in the presence of morin (0 μM – 150 μM). Absorbances at 590 nm for all the sets at different time intervals were plotted. Data are represented as the mean of three replications with $\pm$ SEM. (B) 10^6 cells of *V. cholerae* N16961 were grown in the presence of morin (50 μM – 500 μM) and the Log₁₀ CFU/ml from each set at different time intervals were plotted as a function of time to monitor the killing kinetics. Data are represented as the mean of three replications with $\pm$ SEM.

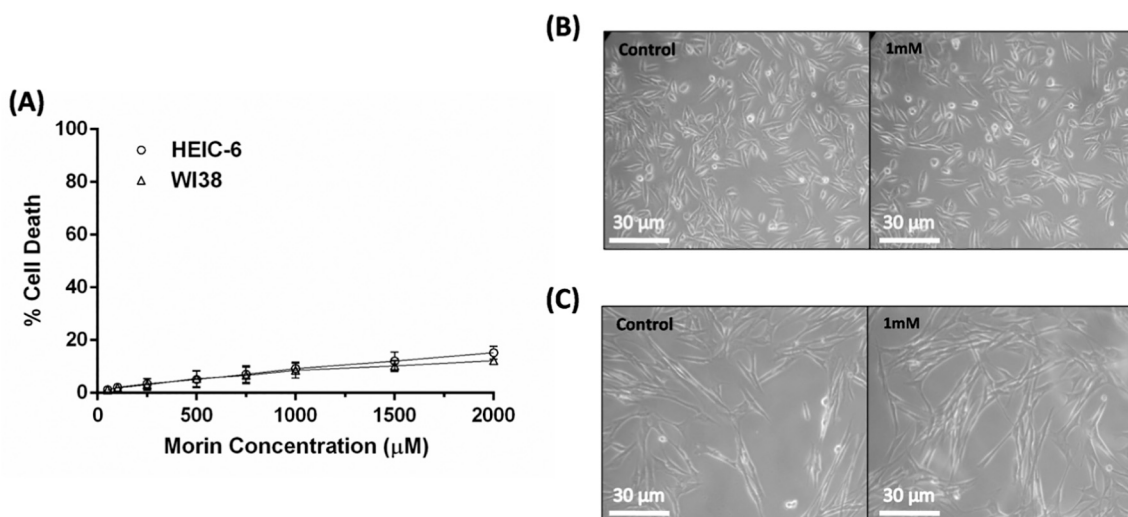


Fig. 2. Effect of morin on WI38 and HIEC-6 cells. (A) Cytotoxic effect of varying doses (0–2000 μM) of morin on WI38 cells and HIEC-6 cells upon 24 h treatment. Cell viability was assessed by the MTT assay, and data are represented as means of three replications with $\pm$ SEM. (B) Bright-field microscopic images of HIEC-6 cells in the absence and in the presence of 1 mM morin. (C) Bright-field microscopic images of WI38 cells in the absence and in the presence of 1 mM morin. Scale bar is 30 μm.

presence of 50 μM morin. Cell lengths were calculated using the ImageJ software by processing ten field-view images for each set of experiments. Elongation of the bacterial cells indicated a defect in the bacterial cytokinesis process, brought about by the presence of morin. The process of cytokinesis in bacteria is governed mainly by the bacterial divisome complex. VcFtsA (FtsA protein of *V. cholerae*) and VcFtsZ (FtsZ protein of *V. cholerae*) being the most important divisome proteins, we were interested to see the effect of morin on the polymerization properties of them.

3.4. Morin inhibits the polymerization of VcFtsA in-vitro

Assembly kinetics of both VcFtsA and VcFtsZ was monitored in the presence of different concentrations of morin (0–40 μM) by a 90° light scattering assay using a fluorimeter keeping excitation and emission wavelength at 550 nm at 37 °C [7,18]. Morin inhibited the polymerization of VcFtsA in a concentration-dependent manner (Fig. 4 A). The inhibition of VcFtsA assembly was found to be maximum at 25 μM morin concentration and was determined to be $72 \pm 7\%$. Further, the effect of morin on VcFtsZ assembly was analyzed. Unlike VcFtsA, no discernable inhibitory effect on the polymerization of VcFtsZ was observed in the presence of increasing concentrations of morin (Fig. 4 B). It could be

inferred from these observations that morin specifically interacted with VcFtsA and not VcFtsZ at the concentrations of morin used in experiments.

3.5. ATP hydrolysis of VcFtsA is affected by morin

We have reported earlier that VcFtsA hydrolyzes ATP and its polymerization is ATP hydrolysis dependent [7]. Owing to the fact that morin inhibits VcFtsA polymerization (Fig. 4A), we were interested in investigating the effect of morin on ATP hydrolysis of VcFtsA. The ATPase activity was found to decrease in the presence of morin in a dose-dependent manner (Fig. 4 C). The rate of ATP hydrolysis by VcFtsA in absence of morin was found to be 2.51 ± 0.11 mol of Pi released/mol of VcFtsA/min. The ATP hydrolysis rate decreased to 1.32 ± 0.09 , 0.83 ± 0.06 , 0.62 ± 0.04 and 0.51 ± 0.03 mol of Pi released/mol of VcFtsA/min in the presence of 5, 10, 15 and 20 μM morin, respectively. From here, it can be hypothesized that morin interacts with VcFtsA by reducing its ATPase activity and thus inhibiting the assembly kinetics of the protein.

3.6. Interaction studies of VcFtsA and morin

VcFtsA has an important role in the bacterial divisome complex. As

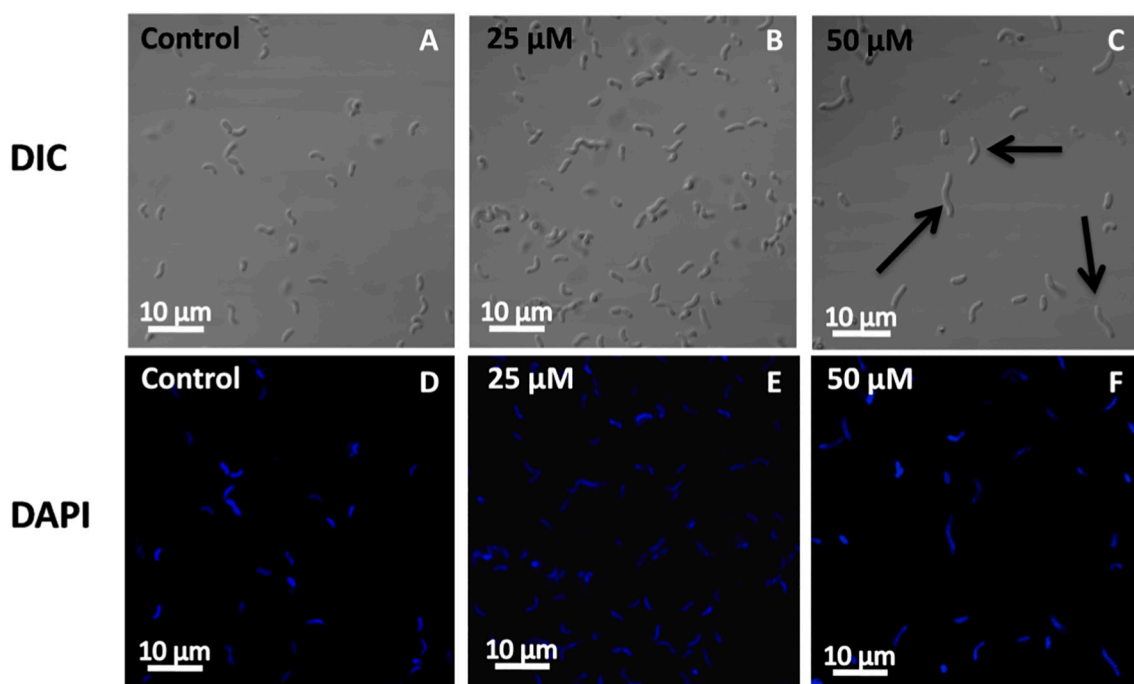


Fig. 3. Effect of morin on the morphology of *V. cholerae* N1696 cells. Elongation of cells was observed with an increasing dose of morin. Differential Interference Contrast (DIC) images of *V. cholerae* N1696 cells in the absence (A) and in the presence of 25 μ M (B) and 50 μ M (C) morin treatment for 4 h. Images (D, E, and F) represent the DAPI stained fluorescence images of images (A, B and C) respectively. Images were obtained by an Olympus Confocal microscope. The scale bars equal to 10 μ m.

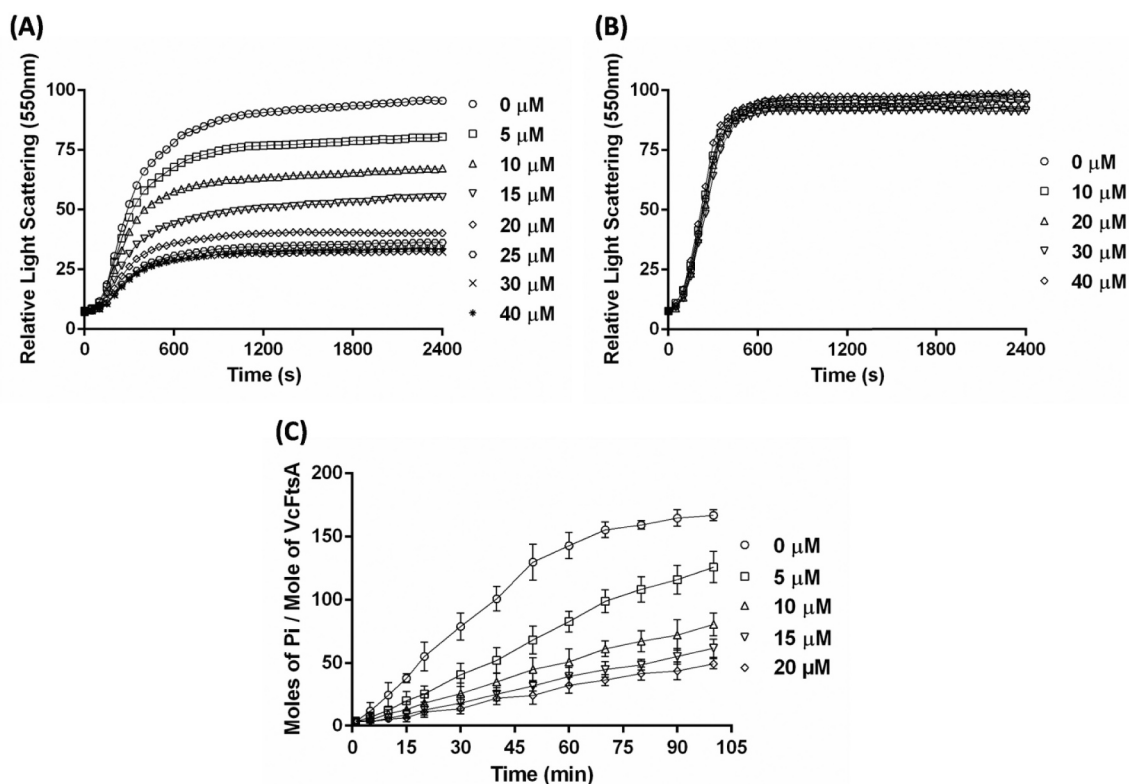


Fig. 4. Effect of morin on the functional properties of VcFtsA and VcFtsZ. (A) 6 μ M VcFtsA was polymerized in 25 mM PIPES (pH 7.0), 10 mM $MgCl_2$, 5 mM $CaCl_2$ and 1 mM ATP in the presence of 0 μ M, 5 μ M, 10 μ M, 15 μ M, 20 μ M, 25 μ M, 30 μ M and 40 μ M of morin, and monitored fluorimetrically over time. (B) 6 μ M VcFtsZ was polymerized in 25 mM PIPES (pH 7.0), 10 mM $MgCl_2$, 10 mM $CaCl_2$ and 1 mM GTP in the presence of 0 μ M, 10 μ M, 20 μ M, 30 μ M and 40 μ M of morin, and monitored fluorimetrically over time. (C) VcFtsA (6 μ M) was incubated in the presence of 25 mM PIPES (pH 7.0), 10 mM $MgCl_2$, 5 mM $CaCl_2$ and 1 mM ATP at 37 $^{\circ}$ C with 0 μ M, 5 μ M, 10 μ M, 15 μ M, 20 μ M, 25 μ M, 30 μ M and 40 μ M of morin concentrations and Pi released was monitored at different time intervals. The data are mean of three independent experiments $\pm$ SEM.

morin inhibited both the polymerization and ATPase activity of the protein, we were interested to study the interaction between VcFtsA and morin. The interaction between VcFtsA and morin was studied by measuring the change of intrinsic emission fluorescence of VcFtsA by exciting its tryptophan residues at 295 nm (Fig. 5 A). Morin has negligible absorbance at 330 nm making the tryptophan residues of VcFtsA as the sole contributors to fluorescence emission recorded. VcFtsA (1 μ M), when incubated with morin (0 μ M – 30 μ M), a fluorescence quenching of the tryptophan residues of VcFtsA was observed, which reached a near-saturation at 30 μ M morin (1:30 M ratio of VcFtsA:Morin) after quenching the fluorescence by nearly 60% (Fig. 5 A). The binding stoichiometry of the VcFtsA – morin interaction was calculated to be 1:1 from the Job Plot (Fig. 5 B).

The interaction between VcFtsA and morin was also monitored using the Far-UV CD technique in phosphate buffer (pH 6.9). CD spectra of 1.5 μ M VcFtsA were taken and a spectrum (mdeg values) was plotted (Fig. 6). Next, morin (2 μ M, 10 μ M, 30 μ M) was added into 1.5 μ M of VcFtsA and the CD spectrums were taken. The CD spectrums (mdeg values) of VcFtsA – morin complexes showed an increased negative ellipticity at both 222 nm and 208 nm wavelength of light, compared to the spectrums (mdeg values) of the individual VcFtsA protein (Fig. 6). From the Far-UV CD spectrums, the total α -helix percent was found for VcFtsA and VcFtsA – morin complexes using the K2D2 server [38]. VcFtsA had a total α -helix percent of $25.9 \pm 0.8\%$, whereas, the total α -helix percent of the VcFtsA – morin complex at 30 μ M morin concentration was $28.6 \pm 0.7\%$. The increase of the α -helix percentage was about 3%, indicating a conformational change in VcFtsA when bound to morin.

3.7. Binding parameters and thermodynamic parameters of VcFtsA-morin interaction

The binding affinity of VcFtsA – morin interaction was obtained by titrating morin (0 μ M – 30 μ M) in a fixed concentration of VcFtsA (1 μ M) at 30 °C. The fluorescence quenching with increasing morin concentrations was observed (Fig. 5 A). The amount of quenching in tryptophan fluorescence at 330 nm wavelength was plotted against different doses of morin (Fig. 7 A). This obtained data were then fitted with a quadratic single-site binding equation using GraphPad Prism software and the dissociation constant (K_d) was determined to be $4.98 \pm 0.71 \mu$ M ($p < 0.01$) (Fig. 7 A). A double-reciprocal plot and a Scatchard plot were also analyzed with the fluorescence quenching data of VcFtsA – morin interaction (Fig. 7 A and B). The K_d value from the double-reciprocal plot was calculated and it was found to be $4.42 \pm 0.73 \mu$ M ($p < 0.01$) (Fig. 7 B) while the same calculated from the Scatchard plot was $4.68 \pm 0.4 \mu$ M ($p < 0.01$) (Fig. 7 C). The three dissociation constant (K_d) values thus obtained are in good agreement with each other. Also, the binding stoichiometry (n) found from the Scatchard Plot was equal to 1.04 ± 0.2 .

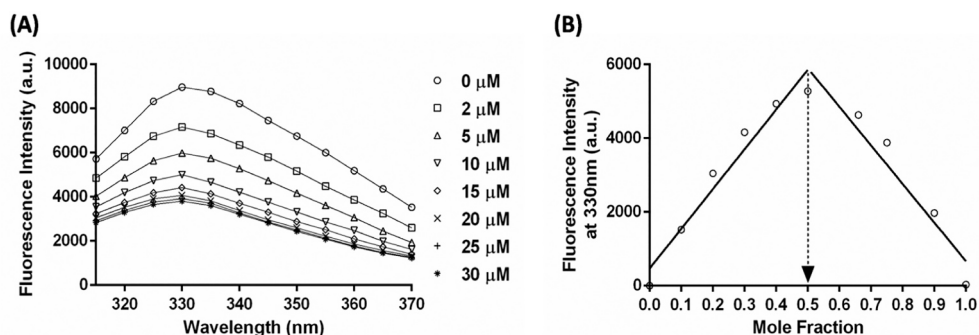


Fig. 5. Interaction study of VcFtsA and morin assessed by fluorescence spectroscopy. (A) Fluorescence of 1 μ M VcFtsA measured in the absence and in presence of different concentrations of morin (0 μ M – 30 μ M). (B) Job Plot for binding of VcFtsA to VcFtsA. Data represent one of three similar experiments (Details of the method are described in “Materials and method section”).

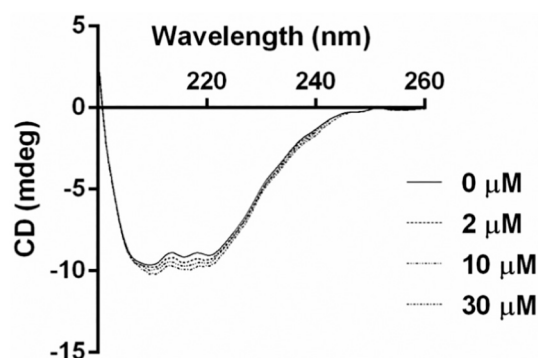


Fig. 6. Interaction study of VcFtsA and morin assessed by circular dichroism. Far-UV CD spectra of 1.5 μ M VcFtsA and complexes of 2 μ M, 10 μ M and 30 μ M morin added to 1.5 μ M VcFtsA. The experiment was performed in 10 mM phosphate buffer (pH, 6.9) at 30 °C.

Likewise, the K_d value of the VcFtsA – morin complex was measured at different temperatures and a Van't Hoff plot was used to analyze the results (Fig. 7 D). From the slope of the plot, the change in enthalpy (ΔH) of VcFtsA – morin interaction was calculated, which was found to be $4.42 \pm 0.61 \text{ kcal mol}^{-1}$. The change in Gibbs free energy (ΔG) value was calculated to be $(-)$ $7.42 \pm 0.37 \text{ kcal mol}^{-1}$ using the K_d value at 30 °C. Finally, the change in entropy (ΔS) was calculated to be $(+)$ $39.07 \pm 0.13 \text{ cal K}^{-1} \text{ mol}^{-1}$ at 30 °C.

3.8. In-silico docking of Morin into VcFtsA

The best 3D model of VcFtsA was obtained using 3wt0.1.A.pdb, which is the template of cell division protein, FtsA, from the SWISS-MODEL template library. The superimposed 3D structure of the template and the modelled VcFtsA protein is shown in Supplementary Fig. S1. The modelled VcFtsA protein was a monomer having a QMEAN value of -1.93 and a GMQE value of 0.61 .

The 3D interaction structure of docked morin with modelled VcFtsA is shown in Fig. 8 A. The pose of morin with which it is bound to VcFtsA is shown in Fig. 8 B. The binding pocket had a centre of (5.77, -9.16 , 60.43) on VcFtsA. The amino acids close to the morin binding sites could be identified also with the ones forming bonds with morin (Fig. 8 C). The pocket consists of hydrophobic amino acid residues like Ala18, Ala213, Ala338, Val259, Ile341, and Tyr375. The polar amino acids present are Gln340, Thr19, and Ser40. Also, Lys258 present at the site is positively charged, whereas Glu373 is negatively charged. The surface of the binding pocket is slightly basic (Fig. 8 D). Though the surface is mostly hydrophilic, some hydrophobic regions are also present (Fig. 8 E). The surface consists of both H-bond donor and acceptor regions (Fig. 8 F). The Gln340 and Ser40 residues of VcFtsA formed hydrogen bond with

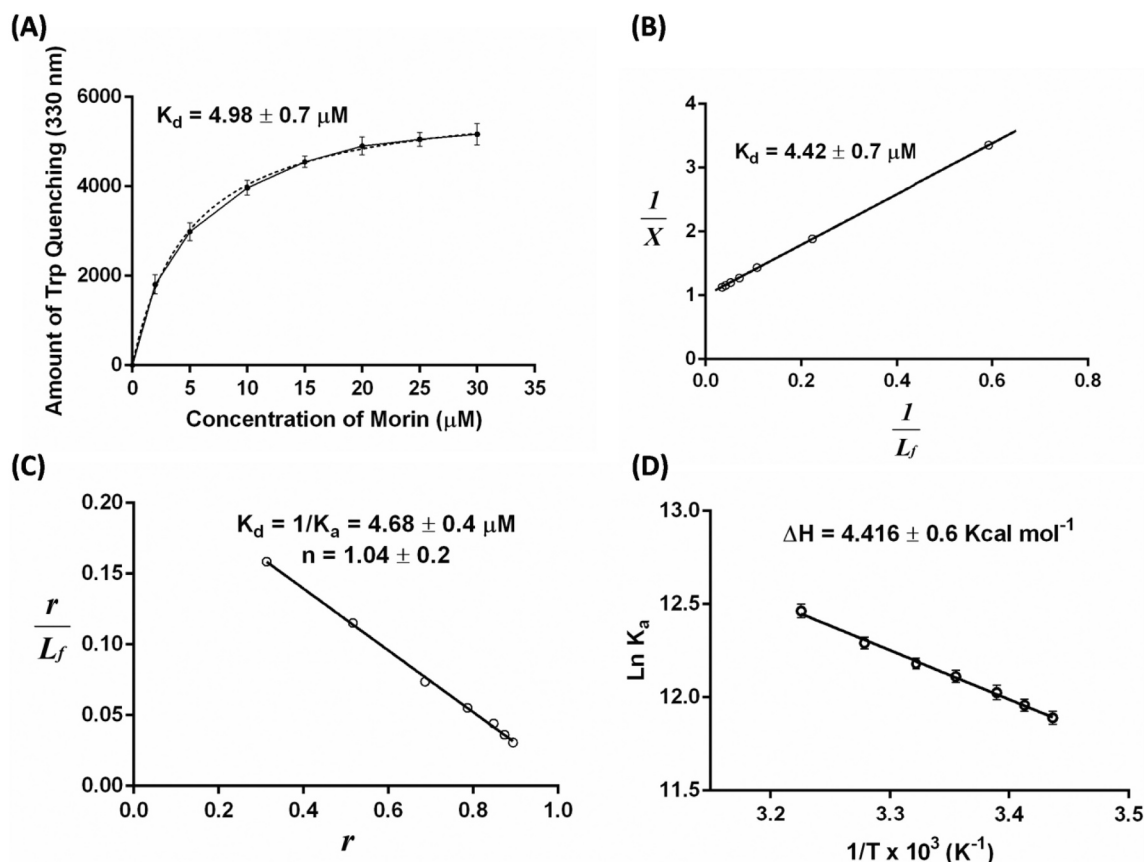


Fig. 7. Interaction study of VcFtsA and morin and Binding Parameters assessed by fluorescence spectroscopy. (A) Tryptophan fluorescence enhancement of VcFtsA ($1 \mu\text{M}$) in the presence of morin ($0 \mu\text{M}$ – $30 \mu\text{M}$) was plotted. The curve showing a change in tryptophan emission (bold line) was fitted with the quadratic equation of single-site binding (dotted line) to find K_d of the interaction. K_d is represented as mean $\pm$ SD ($p < 0.01$). (B) Double-reciprocal plot for binding of morin to VcFtsA. Data are representative of three identical experiments. K_d is represented as mean $\pm$ SD ($p < 0.01$). (C) Scatchard Plot for the interaction between VcFtsA and morin. Data are representative of three identical experiments. K_d is represented as mean $\pm$ SD ($p < 0.01$). (D) Effect of temperature on the equilibrium constant of VcFtsA-morin binding. A Van't Hoff plot for the interaction of VcFtsA with morin is shown with $1 \mu\text{M}$ VcFtsA in the presence of a variable concentration of (0 – 30) μM of morin. The data of the Van't Hoff plot are mean of three independent experiments. ΔH is represented as mean $\pm$ SD ($p < 0.01$).

morin providing the major binding forces. Ser40 bonds with the OH- of C4' located in the B-ring of morin ($\text{O} - \text{H} \cdots \text{O}$ bond), whereas, Gln340 bonds with the O1 from the C-ring of morin ($\text{N} - \text{H} \cdots \text{O}$ bond). Since, morin is rich with pi-electrons, there was formation of π -anion, π -alkyl, π - π T-shaped, and amide- π stacked interactions (Fig. 8 C). All these contributed to the binding affinity of ($-$) $7.2 \text{ kcal mol}^{-1}$.

ATP is known to bind to FtsA through a conserved motif 'VLTGG' [39]. We examined the 3D crystalline structures, 3WQU.pdb (*Staphylococcus aureus* FtsA bound with ATP) and 1E4G.pdb (*Thermotoga maritima* FtsA bound with ATP), and found ATP is bound to the above said motif. In VcFtsA this motif ranges from the amino acids 333 to 337. The structure of docked morin with the modelled VcFtsA (Fig. 8 C) shows Gly337 and its nearby amino acids from VcFtsA are interacting with morin. This suggests that morin and ATP binding sites may have overlapping portions in VcFtsA. The 3D interaction structure of docked ATP with modelled VcFtsA is shown in Fig. 9 A. Also the favourable docking pose of ATP into VcFtsA showed a direct hydrogen bond interaction of ATP with the Gln340 residue of VcFtsA (Supplementary Fig. S2). The binding pocket had a centre of (5.10, -11.64 , 57.61). The binding affinity of was ($-$) $6.8 \text{ kcal mol}^{-1}$. The binding pockets of morin and ATP were hypothetically overlapped on the modelled VcFtsA protein molecule. The centre of binding site location was (5.77, -9.16 , 60.43) for morin and (5.10, -11.64 , 57.61) for ATP. The centre-to-centre distance was $3.81 \text{ \AA}$. But, the radius of binding site spheres of ATP and morin was $9.5 \text{ \AA}$ and $6.6 \text{ \AA}$ respectively. Thus, morin and ATP binds at sites very close to each other and there is a steric hindrance between them (Fig. 9

C). Thus, morin and ATP must compete with each other for binding to VcFtsA. Since, binding affinity of morin ($-7.2 \text{ kcal mol}^{-1}$) is higher than that of ATP ($-6.8 \text{ kcal mol}^{-1}$), binding of morin is energetically favourable than ATP.

It can thus be stated from the docking studies that for the VcFtsA – morin complex, the hydrogen bond involving Gln340 is at the ATP binding site while that formed by Ser40 is away. To experimentally verify these facts, a VcFtsA mutant (VcFtsA-S40A) was cloned and purified. We did not mutate the Gln340 residue as it could alter the ATP binding to VcFtsA.

3.9. Purification and characterization of VcFtsA-S40A

As discussed in the previous section, the amino acid Ser40, and not Gln340, was chosen to be mutated. For this purpose, the 40th amino acid, serine was mutated to alanine and VcFtsA-S40A was cloned using PCR based site-directed mutagenesis [36,37], expressed and purified. After purification, the structural and functional properties of VcFtsA and VcFtsA-S40A were compared. The assembly kinetics of both the proteins were similar (Fig. 10 A). VcFtsA-S40A could hydrolyze ATP and the rate of Pi released by it was highly comparable to that of VcFtsA (Fig. 10 B). Moreover, both the proteins had near-identical fluorescence and CD patterns (Fig. 10 C & D). From these results, it might be concluded that VcFtsA-S40A did not lose the functional and structural properties.

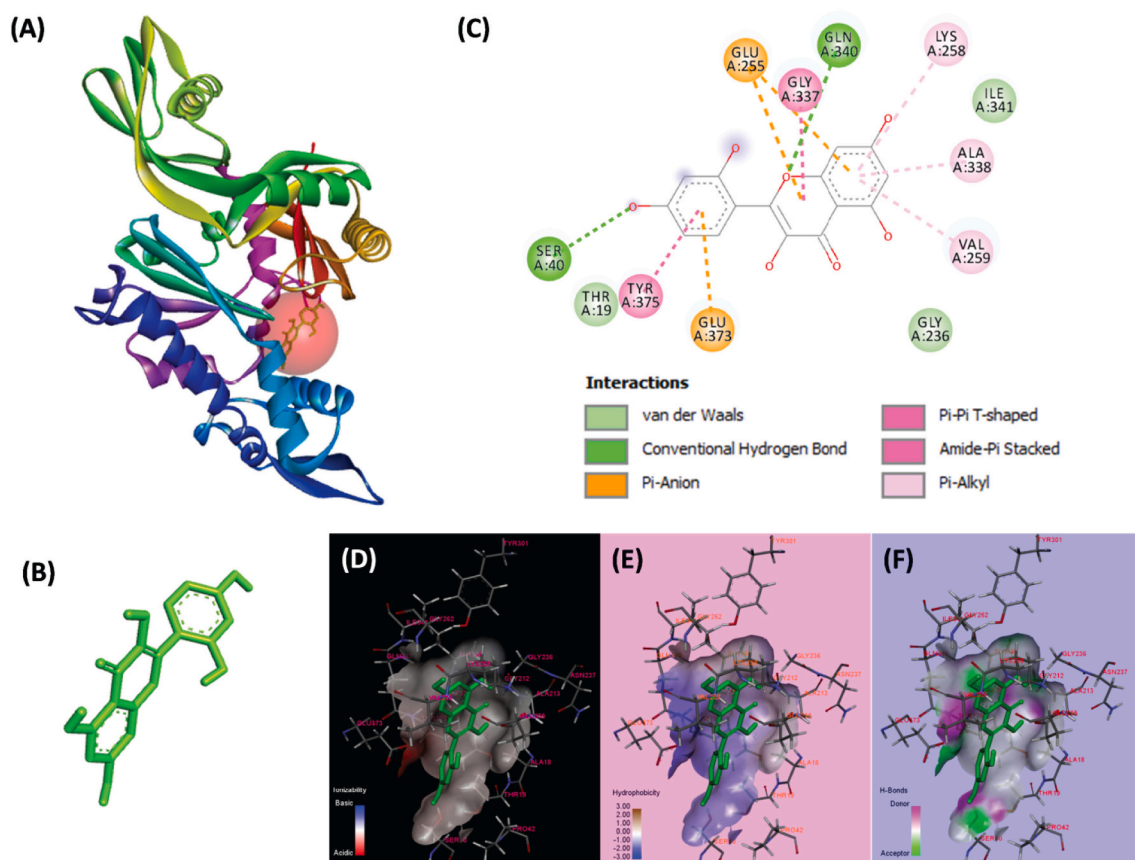


Fig. 8. Interaction structure of docked morin with modelled VcFtsA. (A) The 3D structure of docking. Morin is shown in green and the morin binding-sphere is shown in transparent red. (B) The 3D pose of morin. (C) 2D diagram of the interaction between morin and amino acid residues of VcFtsA involved in bonding. Mapping of the binding site surface; (D) according to ionizability, (E) according to hydrophobicity, and (F) according to H-bond donor-acceptor.

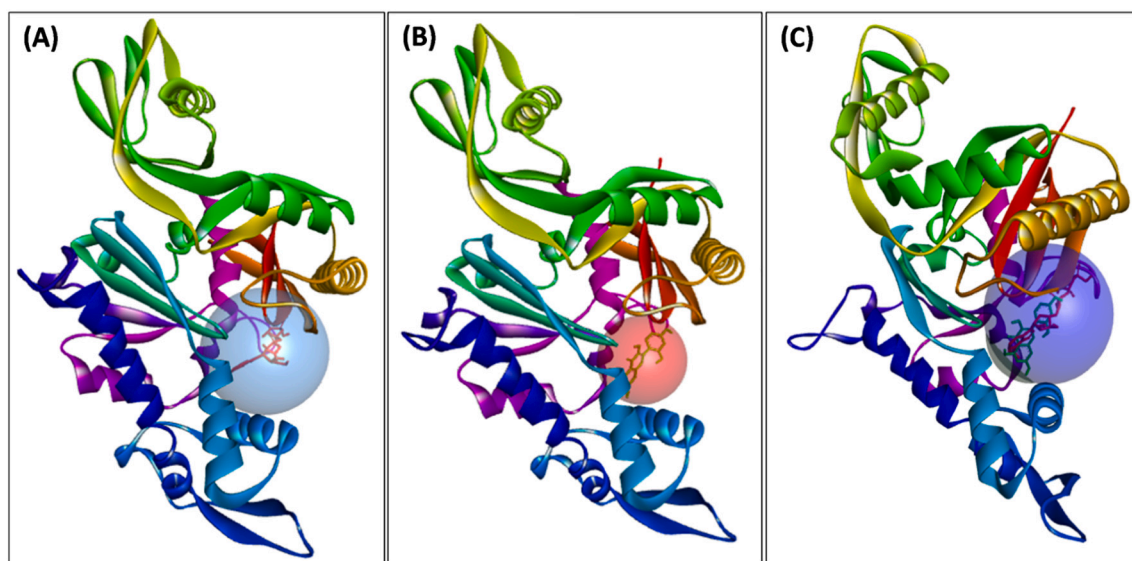


Fig. 9. Interaction structure of docked ATP and morin with modelled VcFtsA. (A) The 3D structure of ATP docked with VcFtsA. ATP is shown in red and the ATP binding-sphere is shown in light blue. (B) The 3D structure of morin docked with VcFtsA. Morin is shown in green and the morin binding-sphere is shown in red. (C) The 3D structure of ATP and morin simultaneously docked with VcFtsA. ATP is shown in red and Morin is shown in green. The ATP binding-sphere is shown in dark blue and the morin binding-sphere is shown in grey.

3.10. Mutant S40A-VcFtsA did not bind with morin

Morin, in the concentration range 0 μ M to 40 μ M, could not alter the

assembly kinetics of VcFtsA-S40A as no significant inhibition of polymerization was observed (Fig. 11 A). When the ATPase activity of VcFtsA-S40A was observed in the presence of morin (0 μ M to 20 μ M), no

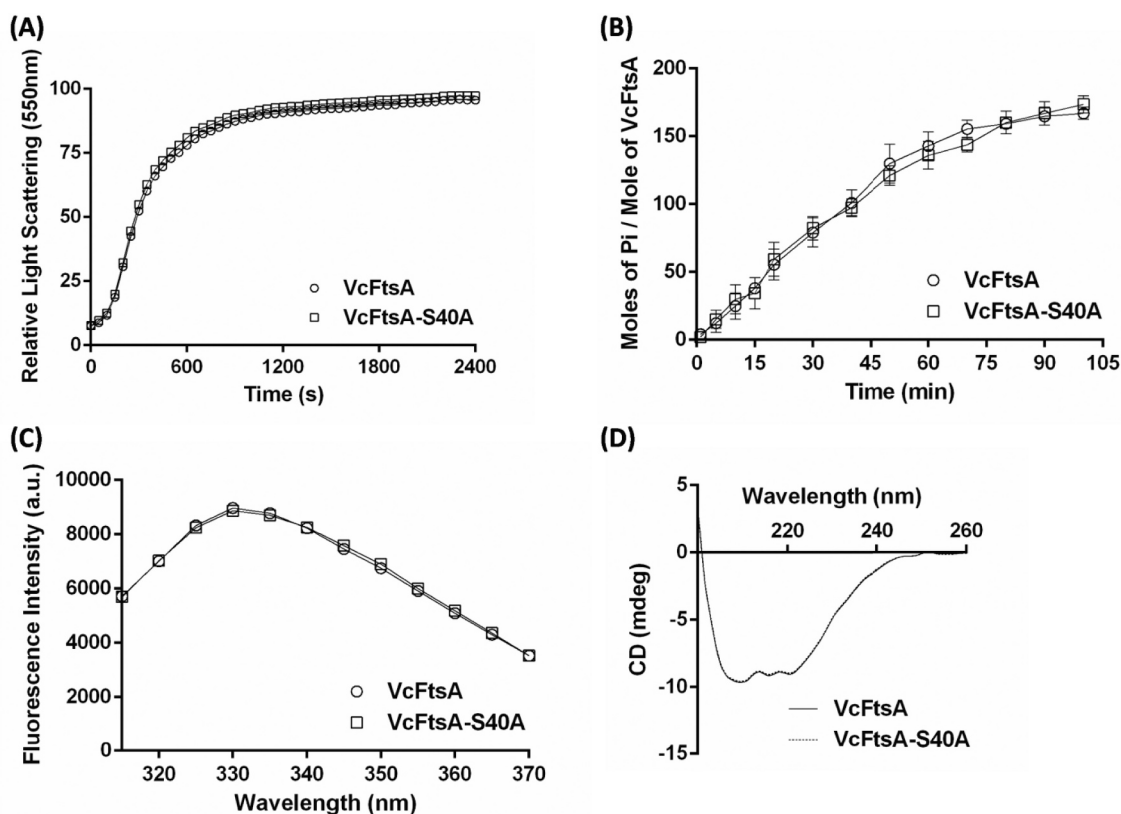


Fig. 10. Comparing the properties of VcFtsA and VcFtsA-S40A. (A) 6 μ M VcFtsA and VcFtsA-S40A were polymerized separately in 25 mM PIPES (pH 7.0), 10 mM MgCl_2 , 5 mM CaCl_2 and 1 mM ATP at 37 $^\circ\text{C}$ and monitored fluorimetrically over time. (B) VcFtsA (6 μ M) and VcFtsA-S40A (6 μ M) were incubated separately in the presence of 25 mM PIPES (pH 7.0), 10 mM MgCl_2 , 5 mM CaCl_2 and 1 mM ATP at 37 $^\circ\text{C}$ and Pi released was monitored at different time intervals. The data plotted are the mean of three independent experiments $\pm$ SEM. (C) Fluorescence spectra of 1 μ M VcFtsA and 1 μ M VcFtsA-S40A when excited at 295 nm wavelength of light. (D) Far-UV CD spectra of 1.5 μ M VcFtsA and 1.5 μ M VcFtsA-S40A. The experiment was performed in 10 mM phosphate buffer (pH, 6.9) at 30 $^\circ\text{C}$.

significant change in the rate of Pi release could be observed (Fig. 11 B). The fluorescence profile of VcFtsA-S40A was also monitored in the presence of morin (0 μ M to 30 μ M), and we failed to observe any noteworthy fluorescence quenching of VcFtsA-S40A in the presence of morin (Fig. 11 C). The secondary structure of VcFtsA-S40A remained unchanged in the presence of morin (0 μ M to 30 μ M) as the ligand was unable to induce any change in the ellipticity values at 222 nm and 208 nm wavelength of light in the CD spectra (Fig. 11 D). Altogether the results indicate that morin binds near Ser40 on FtsA.

Morin binds to VcFtsA with the help of two strong hydrogen bonds. We wanted to find out whether the S40A single mutation is sufficient to abolish the binding of morin to VcFtsA completely. With the help of in-silico docking of morin onto modelled VcFtsA-S40A, we found that after the single mutation, binding of morin to the same site was energetically unfavourable (Supplementary Fig. S4). The binding affinity of morin on VcFtsA-S40A was found to be (+) 2.9 kcal mol⁻¹, whereas its binding affinity on the wild type protein was (-) 7.2 kcal mol⁻¹. This increase in binding affinity was energetically unfavourable for the interaction. Also an unfavourable acceptor-acceptor interaction was present in the VcFtsA-S40A – morin complex. Also in this case no hydrogen bond formed (Supplementary Fig. S4). Thus, clearly the S40A single mutation could abolish the morin VcFtsA binding almost completely.

3.11. Morin could displace Mant-ATP bound to VcFtsA but not from VcFtsA-S40A

Mant-ATP (2'-O-(N-Methylanthraniloyl) Adenosine 5'-Triphosphate) is a fluorescent derivative of ATP that binds to the ATP binding site of proteins. Mant-ATP (500 μ M), when excited at 350 nm, showed a fluorescence emission at 450 nm. Adding VcFtsA (1 μ M) to Mant-ATP (500

μ M) induced a fluorescence emission enhancement and a blue shift (approx 10 nm) in the fluorescence spectrum of Mant-ATP. First, VcFtsA (1 μ M) was added to Mant-ATP (500 μ M) to form a VcFtsA – Mant-ATP complex, then, morin (0 μ M to 30 μ M) was titrated into the complex. It was observed that the fluorescence of VcFtsA – Mant-ATP complex quenched with increasing morin concentrations (Fig. 12 A), indicating that morin could displace Mant-ATP from its complex with VcFtsA. The interaction kinetics of ATP and Mant-ATP with VcFtsA has already been reported earlier [7,40]. VcFtsA – ATP and VcFtsA – Mant-ATP interactions have K_d values of 159.9 ± 8.2 μ M and 305.4 ± 6.8 μ M respectively as reported earlier [7]. These K_d values are many folds higher than that of VcFtsA – morin interaction. Thus, it may be concluded that the binding sites of morin and ATP have overlapping regions in VcFtsA, while morin, having a higher binding affinity towards VcFtsA, could compete and displace ATP from its bound state with VcFtsA.

We repeated the whole experiment with VcFtsA-S40A. Adding VcFtsA-S40A (1 μ M) to Mant-ATP (500 μ M) also induced a similar fluorescence emission enhancement and a blue shift (approx 10 nm) in the fluorescence spectrum of Mant-ATP. Morin (0 μ M to 30 μ M), on the other hand, when titrated into the VcFtsA-S40A – Mant-ATP complex, could not quench the fluorescence of the complex (Fig. 12 B) suggesting that morin could not displace Mant-ATP from its complex with VcFtsA-S40A. These observations reconfirmed that morin could not bind to VcFtsA-S40A.

Morin (30 μ M), VcFtsA (1 μ M), VcFtsA-S40A (1 μ M) and the VcFtsA – morin complex had insignificant fluorescence emission near to 450 nm wavelength of light and all data were plotted after inner-filter effect corrections.

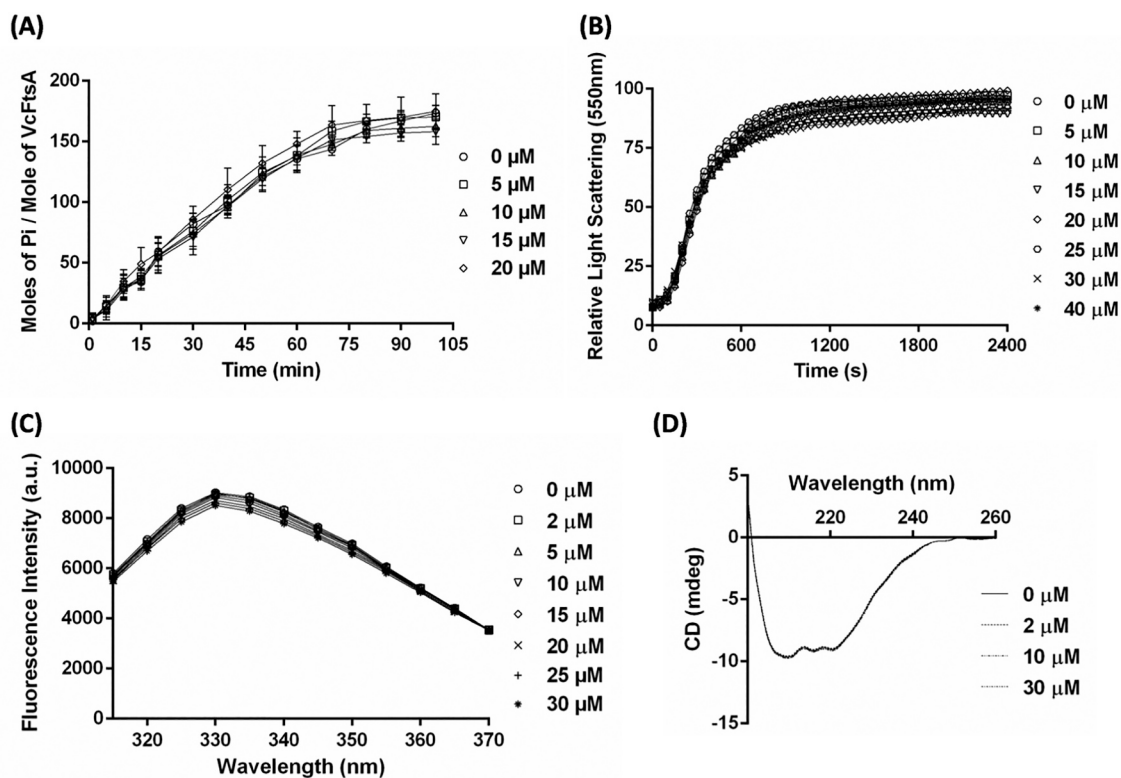


Fig. 11. Interaction study of VcFtsA-S40A and morin. (A) 6 μ M VcFtsA-S40A was polymerized in 25 mM PIPES (pH 7.0), 10 mM $MgCl_2$, 5 mM $CaCl_2$ and 1 mM ATP at 37 $^{\circ}C$ in presence of morin (0–40 μ M) and monitored fluorimetrically over time. (B) VcFtsA-S40A (6 μ M) was incubated in 25 mM PIPES (pH 7.0), 10 mM $MgCl_2$, 5 mM $CaCl_2$ and 1 mM ATP at 37 $^{\circ}C$ in presence of morin (0–20 μ M) and Pi released was monitored at different time intervals. The data plotted are mean of three independent experiments $\pm$ SEM. (C) Fluorescence spectrums of 1 μ M VcFtsA-S40A (excited at 295 nm) in presence of morin (0–30 μ M). (D) Far-UV CD spectra of 1.5 μ M VcFtsA-S40A in presence of morin (0–30 μ M). The experiment was performed in 10 mM phosphate buffer (pH, 6.9) at 30 $^{\circ}C$.

3.12. Morin did not perturb the membrane potential of *V. cholerae* N16961 cells

Finally, we wanted to check whether morin interfered with the membrane potential of *V. cholerae* N16961 cells as a disturbance in the membrane potential also affects cell proliferation [41]. DiOC₂(3), a fluorescent probe, was used for this purpose. DiOC₂(3) exhibits green fluorescence in bacterial cells (Fig. 13). The green fluorescence enhances when the membrane potential is disrupted and shifts to a higher (red) wavelength at higher membrane potentials [34]. Owing to the fact that CCCP, a proton ionophore, is known to disrupt the membrane potential of bacterial cells [35], it is used here as a positive control. Cells treated with CCCP showed good enhancement of the fluorescence spectra of DiOC₂(3). On the other hand, no fluorescence enhancement was observed in cells treated with 50 μ M morin. These observations suggest that morin did not disturb the membrane potential of *V. cholerae* N16961 cells and the elongated morphology of the cells was primarily due to fault in the cell division system induced by morin.

4. Discussion

FtsA is one of the major cell-division proteins essential during the cytokinesis phase of bacterial cell division. It has been reported that FtsA is found in a polymerized form at the mid-cell of bacteria during cell division [42,43]. Polymerized FtsA anchors the Z-ring (formed by polymerized FtsZ protein) to the cell membrane and provides a mechanical force needed for Z-ring shrinkage and effective septum formation [3,44,45]. This makes FtsA a potential target to disrupt the proper cell division in bacteria. In this study, we observed how morin, a naturally occurring flavonoid, interfered with the functional activities of VcFtsA resulting in improper division and lengthening of *V. cholerae*

N16961 cells. Starting with, we first investigated the growth and killing kinetics of *V. cholerae* N16961 cells in the presence of different concentrations of morin (Fig. 1 A & B). The IC₅₀ and MIC values for morin on *V. cholerae* N16961 cells we found to be of 50 μ M and 150 μ M respectively. The MIC value is comparable to previously reported data [8]. Morin is known for its low cytotoxicity on human cell lines [11,12]. Moreover, earlier reports suggest the chemo-protective effect of morin on glomerular mesangial cells [46], neurons [47], cardiovascular cells [48] and hepatocytes [49]. Upon high dose treatment in rats, morin did not show any adverse effect [50]. Likewise, we also observed low cytotoxicity of morin on WI38 and HIEC-6 cells (normal human cell lines). Whereas 500 μ M morin could kill 10⁶ cells of *V. cholerae* N16961 in approximately 3 h, whereas 2 mM concentration of the same could only kill about 20% of both WI38 and HIEC-6 cells in 24 h (Figs. 1 & 2). *V. cholerae* N16961 cells have a comma-shaped morphology, but we observed a four-fold elongated morphology of these cells when incubated with morin at IC₅₀ dose (Fig. 3). Instances of bacterial cell elongation have been reported earlier in other bacterial species as a cause of the faulty functioning of the cell division machinery [4,51–53]. We were interested to examine the effect of morin on the functional properties of two major cell division proteins VcFtsA and VcFtsZ.

Both the proteins were purified by methods discussed earlier⁷. First, trying to investigate the effect of morin on the polymerization property of VcFtsA and VcFtsZ, we found that morin could inhibit the assembly of VcFtsA nearly up to 72% in a concentration-dependent manner. On the other hand, at similar concentrations of morin, the assembly kinetics of VcFtsZ remained unaltered. These observations pointed to the inference that morin, being unable to interact with VcFtsZ polymerization, should have no effect on the Z-ring formation within the cell. The polymerization of VcFtsA at mid-cell, on the other hand, can be affected by the presence of morin. Various reports show that FtsA remains a polymer

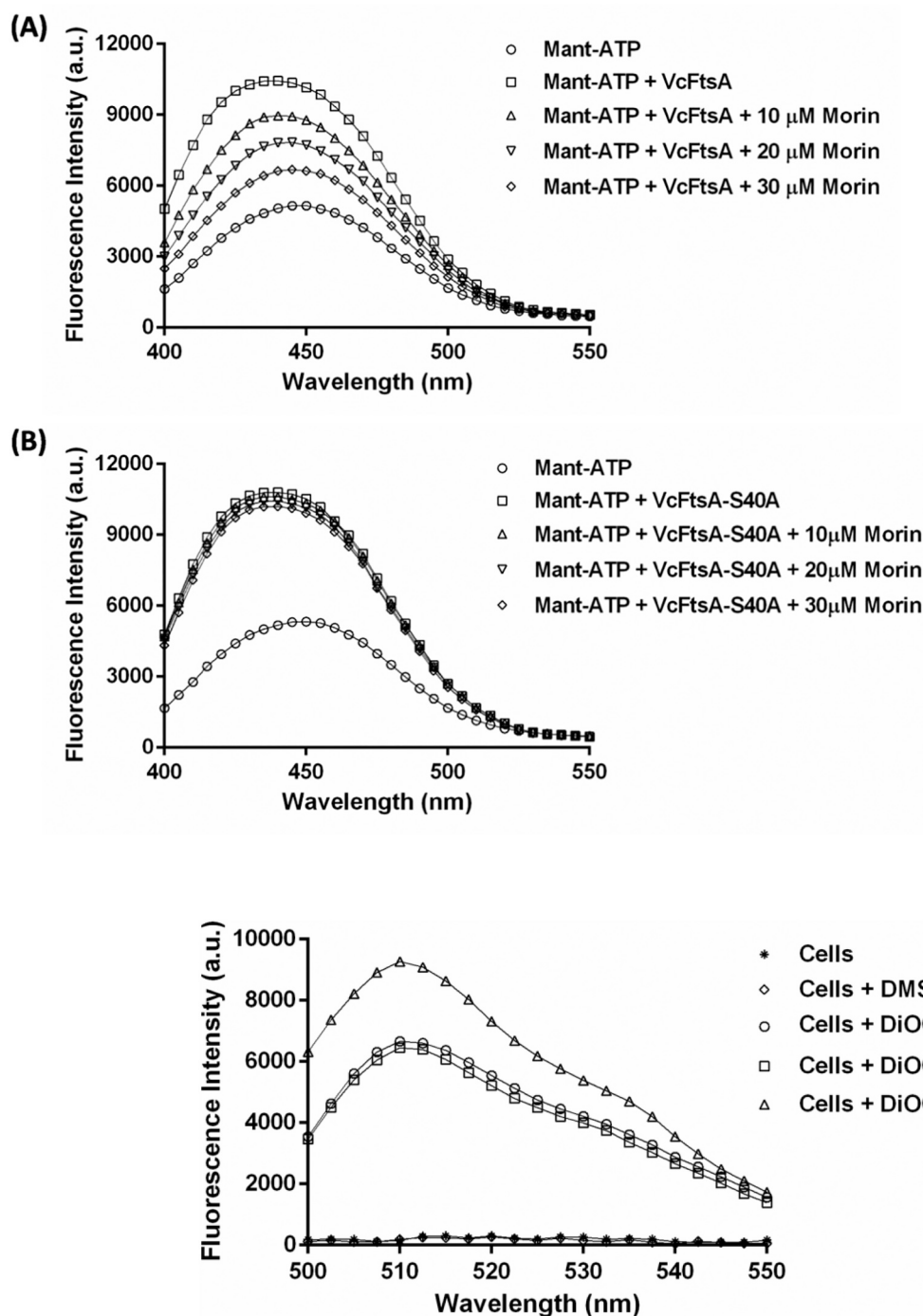


Fig. 12. Morin displaces Mant-ATP from VcFtsA but not from VcFtsA-S40A. (A) The fluorescence of 500 μ M Mant-ATP is increased when 1 μ M VcFtsA is added to it. To the VcFtsA – Mant-ATP complex morin (0–30 μ M) was added and a dose-wise fluorescence quenching was observed. (B) The fluorescence of 500 μ M Mant-ATP is increased when 1 μ M VcFtsA-S40A is added to it. To the VcFtsA-S40A – Mant-ATP complex morin (0–30 μ M) was added but no fluorescence quenching was observed. The plotted data are the best of three separate experimental sets.

Fig. 13. Morin did not perturb the Membrane Potential of *V. cholerae* N16961 cells. Cells alone or when incubated with DMSO showed negligible fluorescence. Cells incubated with 20 μ M DiOC₂(3) is plotted. Enhancement in fluorescence was observed in cells treated with 1 μ M CCCP but not when cells were treated with 50 μ M morin. The plotted data are the best of three separate experimental sets.

between the cell membrane and the Z-ring [43,54] which explains why affecting the polymerization of FtsA can result in faulty cell-division machinery leading to cell elongation. Looking for an answer to how morin affects the assembly of VcFtsA, we found out that the ATPase activity of VcFtsA was hindered due to the presence of morin. It has been reported earlier by us that VcFtsA polymerization is ATP-hydrolysis dependent [7]. Thus by affecting the rate of ATP hydrolysis, morin inhibited VcFtsA polymerization (Fig. 4).

Binding of morin and VcFtsA was studied using fluorimetry. A quenching in the intrinsic fluorescence intensity of VcFtsA suggested interaction between VcFtsA and morin and it was bound to the protein in

a 1:1 stoichiometry (Fig. 5B). The change in the secondary structure of VcFtsA was also evident from the CD spectra (Fig. 6). The dissociation constant (K_d) for VcFtsA and morin interaction was found to be around 4.7 μ M (Fig. 7) from the Scatchard plot, double reciprocal plot and single-binding site plot. The K_d value suggests a moderately strong binding between the protein and the ligand. Thermodynamic studies on the interaction between VcFtsA and morin showed a negative ΔG value indicating a spontaneous binding whereas the positive ΔH value indicates that the interaction is endothermic in nature. The high negative value of $T\Delta S$ compared to the positive enthalpy signified that the interaction is entropy-driven. Although hydrogen bonds between VcFtsA

and morin are a major factor of the binding, the entropy is positive for the interaction meaning that the overall disorderness increases after morin binds with VcFtsA. A most probable reason is that the water structure of VcFtsA is changed after binding with morin, increasing the disorderness.

The *in-silico* docking studies gave us a preliminary idea that the major bonds (hydrogen bonds) between VcFtsA and morin involved the Ser40 and Gln340 amino acids of VcFtsA (Fig. 8). Comparing the 3D model with the known crystal structures of FtsA – ATP led us to hypothesize that the binding site of morin and ATP to VcFtsA should have some overlapping regions with Gln340 very close to the ATP binding site of VcFtsA. Hypothetically setting up the binding sites of morin and ATP on modelled VcFtsA showed both the molecules bind to sites on VcFtsA that are very close to each other and must have a steric hindrance if both tried to bind simultaneously (Fig. 9). The VcFtsA-S40A mutant protein helped us experimentally validate these claims. VcFtsA-S40A had its functional (assembly kinetics and ATPase activity) and structural (fluorescence profile and CD spectrum) properties similar to that of its wild type counterpart (Fig. 10). Due to the mutation in the Ser40 residue, morin could not form any hydrogen bond with Ala40 amino acid. The absence of this hydrogen bond either made it impossible for morin to bind to its binding-pocket in VcFtsA, or the K_d of the new interaction was so high that the VcFtsA-S40A – morin complex was totally unstable at 40 μ M morin concentration (Fig. 11). The fact that morin could not affect the assembly kinetics and ATPase activity of VcFtsA-S40A also confirmed that morin could not bind to the mutant protein and validated the binding site determined by *in-silico* studies. It also added strength to our reasoning that the binding sites of morin and ATP are close to each other in VcFtsA. Finally, when morin was found to compete with Mant-ATP and displace it from its complex with VcFtsA, it was confirmed that the binding sites of morin and ATP had overlapping regions. Morin has a better affinity for VcFtsA as it has a lower K_d than that of ATP with the same protein [7]. This is self-explanatory why morin is privileged to bind to VcFtsA than ATP. Thus, in summary, we can state that the binding of morin to VcFtsA leads to the inability of ATP to bind the protein which, in turn, decreases the rate of ATP hydrolysis by VcFtsA inhibiting its polymerization.

In conclusion, morin a natural flavonoid can bind to VcFtsA inhibiting ATP binding to the protein. ATP hydrolysis, which is needed by the protein to form polymers, is inhibited resulting in altered assembly kinetics of VcFtsA. Now, VcFtsA polymerization, which is essential for the correct functioning of the cell division complex, undergoes alterations in the presence of morin leading to faulty cell-division machinery in *V. cholerae* N16961 cells, which shows an elongated phenotype. We thus elucidated the mode of action of morin against *V. cholerae* N16961 cells. The insight obtained from the VcFtsA – morin binding will help us in the future to design synthetic molecules that can be used to target VcFtsA with better efficacy and screen potent anti-bacterial agents.

Author contributions

DN and GC had major contributions to the conception and design of the study, analysis and interpretation of the data, and writing the manuscript. Acquisition of the data was done by DN. DGD contributed in the *in-silico* study data acquisition.

Credit author statement

DN designed the experiments, performed the experiments, analyzed the data, wrote manuscript. DGD performed the *in-silico* experiments, analyzed the data, and contributed in writing the relevant part of the manuscript. GC conceived the project, designed the experiments, analyzed the data and also wrote the manuscript.

Declaration of Competing Interest

The authors declare that they have no conflicts of interest with the contents of this article.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.bbagen.2021.129931>.

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